

NOTES: CLASS-2

Fourier Series of Even and Odd functions

A function $f(x)$ is said to be even if $f(-x) = f(x) \quad \forall x$ and odd if $f(-x) = -f(x) \quad \forall x$.

Fourier series of an even function $f(x)$ in $(-\pi, \pi)$ contains only cosine terms and is known as Fourier cosine series given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx)$$

Where $a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx$ And $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$

Fourier series of an odd function $f(x)$ in $(-\pi, \pi)$ contains only sine terms and is known as Fourier sine series given by

$$f(x) = \sum_{n=1}^{\infty} (b_n \sin nx) \text{ Where } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

Fourier series of functions having period $2L$

Let $f(x)$ be a periodic function with arbitrary period $2L$ defined in an interval $c < x < c + 2L$.

Then the Fourier series expansion of $f(x)$ is $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$

Where $a_0 = \frac{1}{L} \int_c^{c+2L} f(x) dx$

$$a_n = \frac{1}{L} \int_c^{c+2L} f(x) \cos \left(\frac{n\pi x}{L} \right) dx$$

$$b_n = \frac{1}{L} \int_c^{c+2L} f(x) \sin \left(\frac{n\pi x}{L} \right) dx.$$

Fourier series for even and odd functions defined in $(-L, L)$

Fourier series of an even function $f(x)$ in $(-L, L)$ contains only cosine terms and is known as Fourier cosine series given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} \right)$$

Where $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx = \frac{2}{L} \int_0^L f(x) dx$

And $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$

Fourier series for even and odd functions defined in $(-L, L)$

Fourier series of an odd function $f(x)$ in $(-L, L)$ contains only sine terms and is known as Fourier sine series given by

$$f(x) = \sum_{n=1}^{\infty} \left(b_n \sin \frac{n\pi x}{L} \right)$$

$$\text{Where } b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

Problem 1. Find the Fourier Series expansion of $f(x) = x(1-x)(2-x)$ in $(0,2)$. Deduce the sum of the series $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \dots$

Solution: The Fourier expansion of $f(x)$ over $(0,2l)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

The period of $f(x) = 2 - 0 = 2$. Therefore, $2l = 2 \Rightarrow l = 1$

The relevant Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\pi x + b_n \sin n\pi x)$$

where

$$a_0 = \int_0^2 f(x) dx.$$

$$= \int_0^2 2x - 3x^2 + x^3 dx.$$

$$= \left[2 \frac{x^2}{2} - 3 \frac{x^3}{3} + \frac{x^4}{4} \right]_0^2$$

$$= 4 - 8 + \frac{16}{4}$$

$$\Rightarrow a_0 = 0$$

$$a_n = \int_0^2 f(x) \cos n\pi x dx$$

$$= \int_0^2 (2x - 3x^2 + x^3) \cos n\pi x dx$$

$$= \left[(2x - 3x^2 + x^3) \left(\frac{\sin(n\pi x)}{n\pi} \right) - (2 - 6x + 3x^2) \left(\frac{-\cos(n\pi x)}{(n\pi)^2} \right) + (-6 + 6x) \left(\frac{-\sin(n\pi x)}{(n\pi)^3} \right) - \right.$$

$$\left. (6) \left(\frac{\cos(n\pi x)}{(n\pi)^4} \right) \right]_0^2$$

$$= \left[(2 - 6x + 3x^2) \left(\frac{-\cos(n\pi x)}{(n\pi)^2} \right) + (-6) \left(\frac{\sin(n\pi x)}{(n\pi)^4} \right) \right]_0^2$$

$$= \left[(2) \left(\frac{-\cos(2n\pi)}{(n\pi)^2} \right) - (2) \left(\frac{-1}{(n\pi)^2} \right) \right] + \left[(-6) \left(\frac{\cos(2n\pi)}{(n\pi)^4} \right) - (-6) \left(\frac{1}{(n\pi)^4} \right) \right]$$

$$\Rightarrow a_n = 0$$

$$b_n = \int_0^2 f(x) \sin n\pi x dx$$

$$= \int_0^2 (2x - 3x^2 + x^3) \sin n\pi x dx$$

$$= \left[(2x - 3x^2 + x^3)(-\cos n\pi x) - (2 - 6x + 3x^2) \left(\frac{-\sin(n\pi x)}{(n\pi)^2} \right) + (-6 + 6x) \left(\frac{\cos(n\pi x)}{(n\pi)^3} \right) - (6) \left(\frac{\sin(n\pi x)}{(n\pi)^4} \right) \right]_0^2$$

$$= \left[(2x - 3x^2 + x^3)(-\cos n\pi x) + (-6 + 6x) \left(\frac{\cos(n\pi x)}{(n\pi)^3} \right) \right]_0^2$$

$$= \left[(0)(-\cos 2n\pi) - (0)(-1) \right] + \left[(6) \left(\frac{\cos(2n\pi)}{(n\pi)^3} \right) - (-6) \left(\frac{1}{(n\pi)^3} \right) \right]$$

$$\Rightarrow b_n = \frac{12}{n^3\pi^3}$$

The required Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\pi x + b_n \sin n\pi x)$$

$$\Rightarrow f(x) = \sum_{n=1}^{\infty} \left(\frac{12}{n^3\pi^3} \sin n\pi x \right)$$

Deduction :

To deduce the series we shall put $x = \frac{1}{2}$ in the above equation,

$$f\left(\frac{1}{2}\right) = \sum_{n=1}^{\infty} \left[\frac{12}{n^3\pi^3} \sin n\pi \left(\frac{1}{2}\right) \right]$$

$$\Rightarrow 2\left(\frac{1}{2}\right) - 3\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 = \frac{12}{\pi^3} \sum_{n=1}^{\infty} \left[\frac{1}{n^3} \sin\left(\frac{n\pi}{2}\right) \right]$$

$$\Rightarrow \frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \dots = \frac{\pi^3}{32}$$

REFERENCES:

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